

ETDS QA Report YEARLY

QA Details

Tested by	QMP2, QMP3		
Phantom	SURF	Linac	4
Date	06/08/2026	Time	17:36

Passrates for all 6 DOF

	DOF	pass [%]	watch [%]	action [%]	results
06/08/2026 0.710 Pass M 100%	longitudinal	100	0	0	Pass
	lateral	100	0	0	Pass
	vertical	100	0	0	Pass
	roll	100	0	0	Pass
	pitch	100	0	0	Pass
	yaw	100	0	0	Pass

Maximum error metrics

Max. MAE	Max. RMSE	Max. Absolute Error
0.210 mm	0.294 °	0.710 °

Applied tolerance levels and evaluation criteria

Metric	Pass	Watch	Action
Tracking performance (translation)	≤ 1 mm	1 – 2 mm	> 2 mm
Tracking performance (rotation)	$\leq 1^\circ$	1 – 2°	$> 2^\circ$
Criterion			
Pass rate [%]	≥ 99	95–99	< 95
Watch rate [%]	≤ 1	1–5	–
Action rate [%]	0	0	> 0

Abbreviations:

DOF = degree of freedom

MAE = mean average error

RMSE = root mean square error

Date

Signature (QMP2 / QMP3)

QA Report Details

Description

The aim of the annual quality assurance (QA) measurements is to evaluate the surface scanning accuracy of the ExacTrac Dynamic system using a mechanically movable and heated phantom. Four measurement series are performed, comprising two different deflections: Deflection 1, in which the phantom is displaced in smaller increments (up to 20 mm and 1°), and Deflection 2, with larger increments (up to 50 mm and 2°). Each deflection is carried out with both unheated and heated heating pads. The phantom is displaced according to predefined translational and rotational motion sequences. Before, between, and after each motion sequence, the position of the sphere, located at the center of the phantom, initially aligned with the treatment isocenter, is determined using sphere detection to verify the system's localization accuracy as part of the annual QA procedure. The underlying base kinematics of the phantom is calibrated against the sphere detection system, averaged across all linacs; the dynamic displacement between the targeted positions themselves is driven by stepper motors accurate to the micrometer range and therefore remains independent of this calibration.

Passrates for all measurments

DOF	Deflection	Temp.	pass [%]	watch [%]	action [%]	result
longitudinal	1	32C	100	0	0	Pass
longitudinal	2	32C	100	0	0	Pass
longitudinal	1	RT	100	0	0	Pass
longitudinal	2	RT	100	0	0	Pass
lateral	1	32C	100	0	0	Pass
lateral	2	32C	100	0	0	Pass
lateral	1	RT	100	0	0	Pass
lateral	2	RT	100	0	0	Pass
vertical	1	32C	100	0	0	Pass
vertical	2	32C	100	0	0	Pass
vertical	1	RT	100	0	0	Pass
vertical	2	RT	100	0	0	Pass
roll	1	32C	100	0	0	Pass
roll	2	32C	100	0	0	Pass
roll	1	RT	100	0	0	Pass
roll	2	RT	100	0	0	Pass
pitch	1	32C	100	0	0	Pass
pitch	2	32C	100	0	0	Pass
pitch	1	RT	100	0	0	Pass
pitch	2	RT	100	0	0	Pass
yaw	1	32C	100	0	0	Pass
yaw	2	32C	100	0	0	Pass
yaw	1	RT	100	0	0	Pass
yaw	2	RT	100	0	0	Pass

QA Report Details

Error Metrics						
DOF	Unit	Deflection	Temp.	MAE	RMSE	Max Error
longitudinal	mm	1	32C	0.133	0.165	0.354
longitudinal	mm	2	32C	0.194	0.228	0.496
longitudinal	mm	1	RT	0.142	0.188	0.530
longitudinal	mm	2	RT	0.210	0.260	0.693
lateral	mm	1	32C	0.091	0.124	0.200
lateral	mm	2	32C	0.210	0.258	0.500
lateral	mm	1	RT	0.074	0.104	0.200
lateral	mm	2	RT	0.207	0.255	0.500
vertical	mm	1	32C	0.051	0.068	0.164
vertical	mm	2	32C	0.160	0.197	0.394
vertical	mm	1	RT	0.093	0.120	0.225
vertical	mm	2	RT	0.177	0.223	0.494
roll	°	1	32C	0.038	0.058	0.125
roll	°	2	32C	0.082	0.119	0.297
roll	°	1	RT	0.065	0.077	0.126
roll	°	2	RT	0.079	0.115	0.292
pitch	°	1	32C	0.061	0.073	0.202
pitch	°	2	32C	0.076	0.100	0.248
pitch	°	1	RT	0.065	0.085	0.215
pitch	°	2	RT	0.078	0.096	0.278
yaw	°	1	32C	0.044	0.083	0.251
yaw	°	2	32C	0.182	0.294	0.710
yaw	°	1	RT	0.077	0.113	0.285
yaw	°	2	RT	0.166	0.274	0.697

Maximum error metrics compared with previous years

Year	Max. MAE	Max. RMSE	Max. Absolute Error
2026	0.210 mm	0.294 °	0.710 °

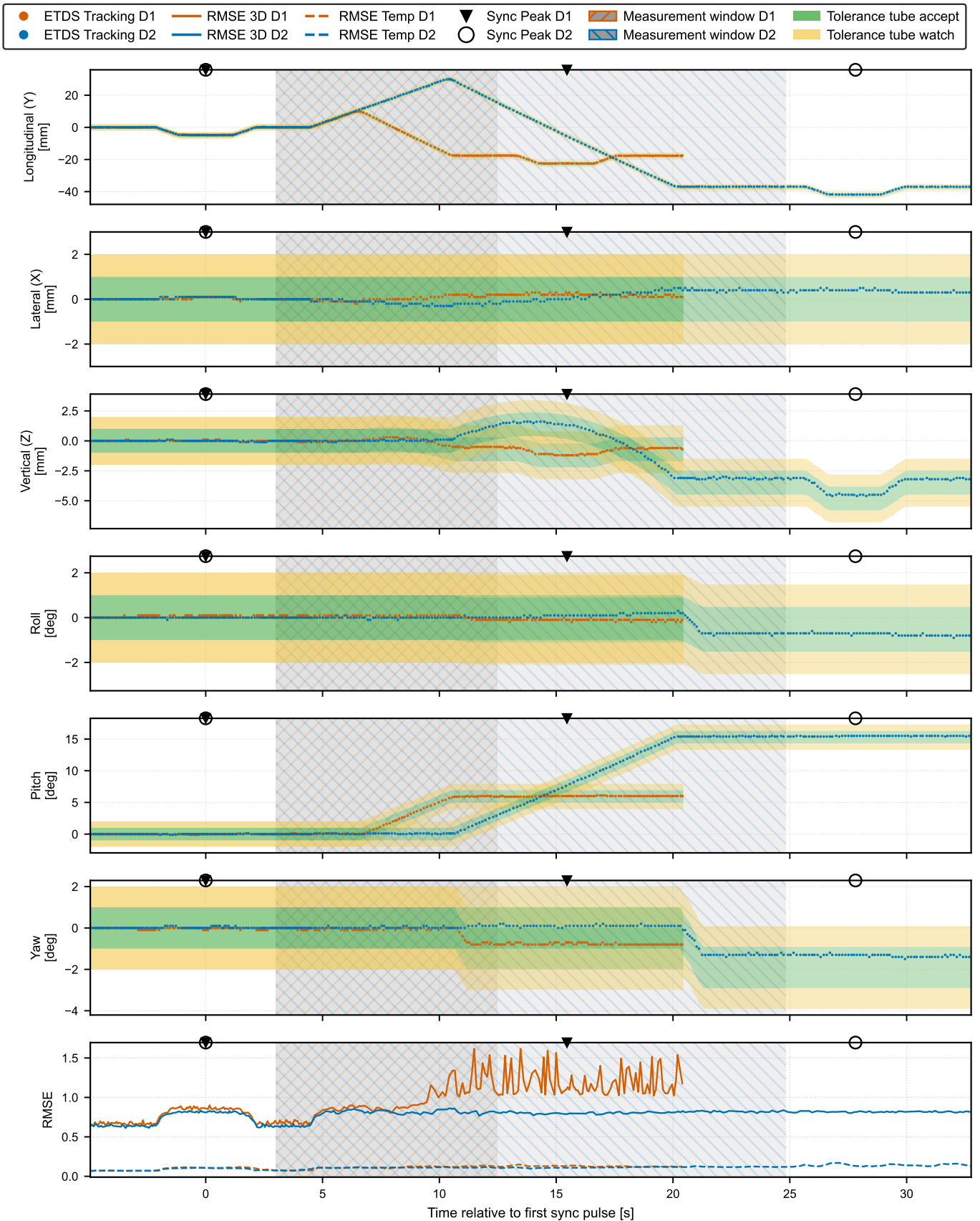
Abbreviations:

DOF = degree of freedom
MAE = mean average error
RMSE = root mean square error
SD = sphere detection

Contact:

<https://github.com/AG-Kollotzek/surf-etds-qa/issues>

ETDS QA Overview - L4 | Pads RT°C | Couch 0deg



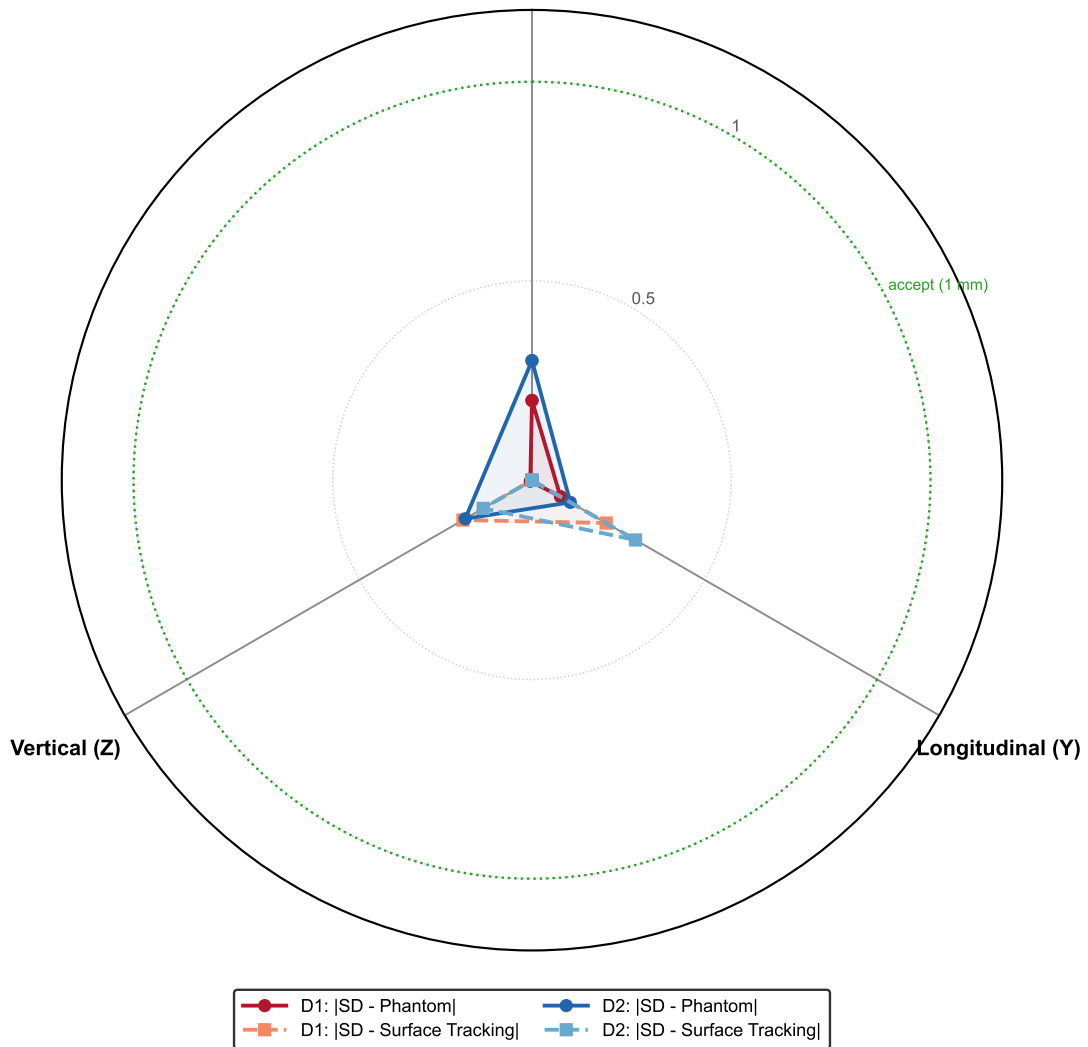
Sphere detection comparison - vectors as (x, y, z) in mm

Defl.	SD Reference Coordinates	SD Movement Vector	Phantom Movement Vector	Movement Error (SD - Phantom)	Surface Tracking Vector
1	(-1.10, -0.60, -0.30)	(0.20, -17.80, -0.70)	(0.00, -17.72, -0.69)	(0.20, -0.08, -0.01)	(0.20, -17.58, -0.50)
2	(-1.10, -0.50, -0.30)	(0.30, -37.20, -3.30)	(0.00, -37.31, -3.49)	(0.30, 0.11, 0.19)	(0.30, -36.90, -3.16)

Sphere detection - absolute deviations

ETDS QA Overview - L4 | Pads RT°C | Couch 0deg

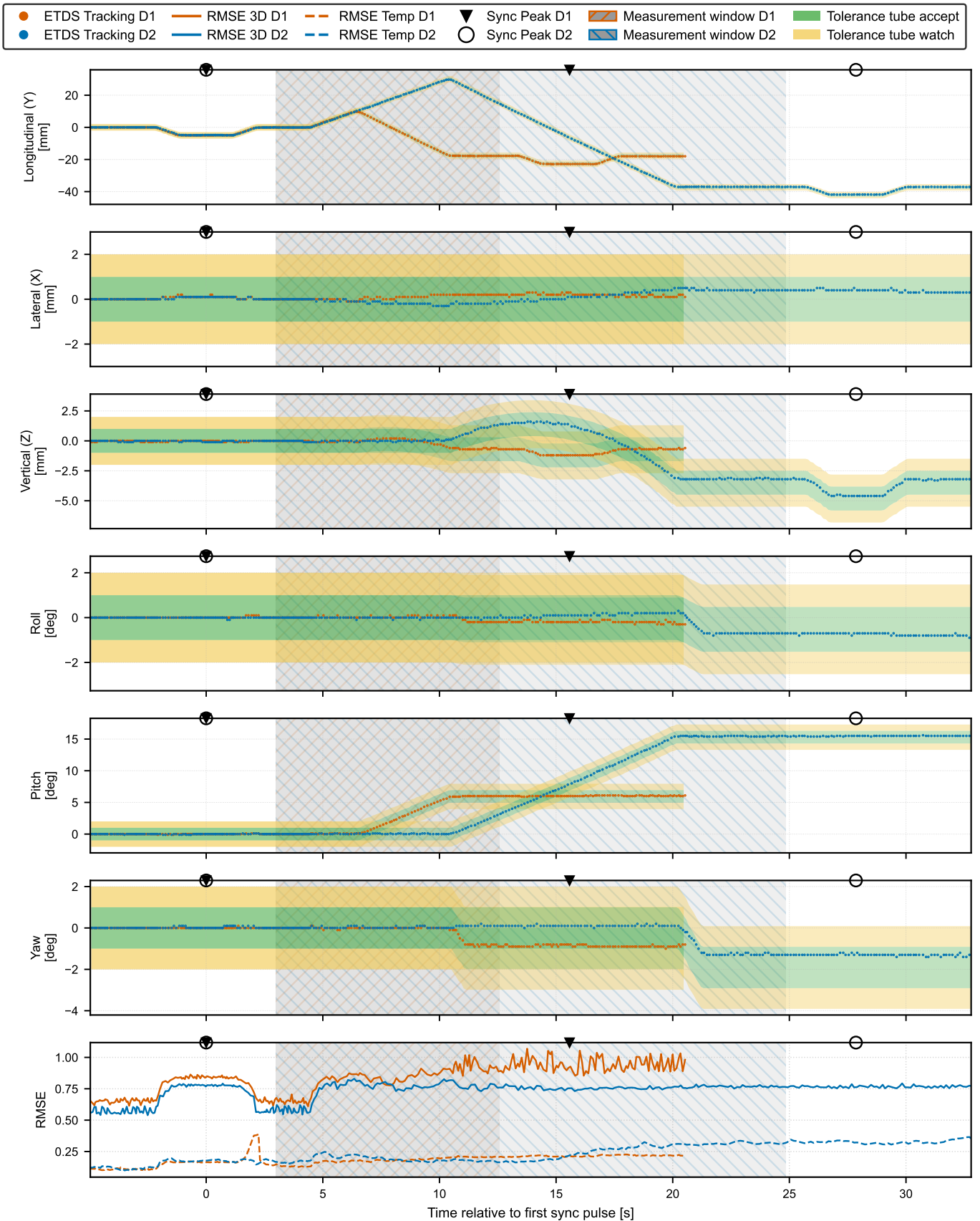
Lateral (X)



Defl.	Deviation	Lateral (X) [mm]	Longitudinal (Y) [mm]	Vertical (Z) [mm]
D1	SD - Phantom	0.20	0.08	0.01
D1	SD - Surface Tracking	0.00	0.22	0.20
D2	SD - Phantom	0.30	0.11	0.19
D2	SD - Surface Tracking	0.00	0.30	0.14

Absolute deviations of the X-ray based sphere detection from the phantom axis setpoint and from the ETD surface tracking, evaluated at the deflected position (end of measurement window). Tolerance rings: accept 1 mm, watch 2 mm.

ETDS QA Overview - L4 | Pads 32°C | Couch 0deg



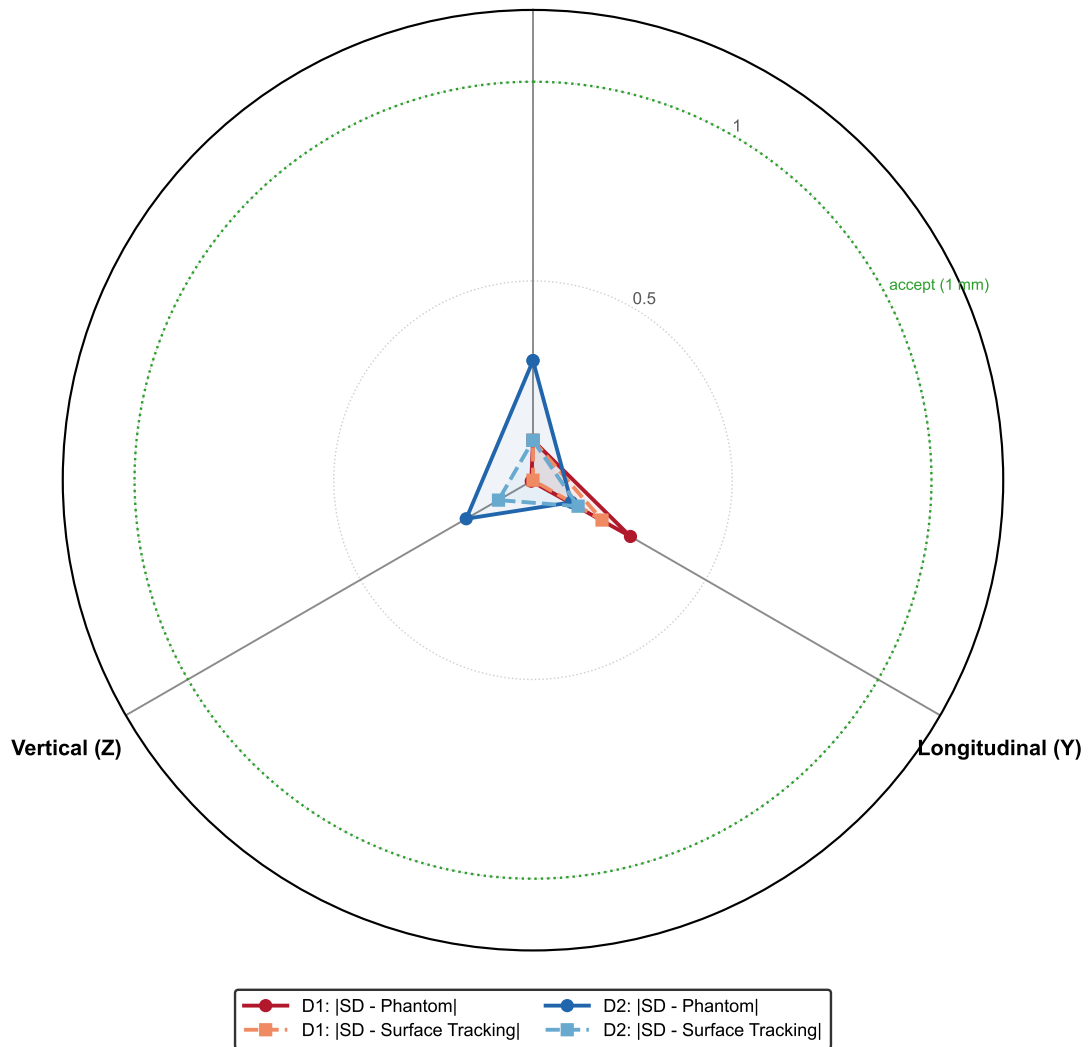
Sphere detection comparison - vectors as (x, y, z) in mm

Defl.	SD Reference Coordinates	SD Movement Vector	Phantom Movement Vector	Movement Error (SD - Phantom)	Surface Tracking Vector
1	(-1.00, -0.30, -0.20)	(0.10, -18.00, -0.70)	(0.00, -17.72, -0.69)	(0.10, -0.28, -0.01)	(0.20, -17.80, -0.70)
2	(-1.10, -0.50, -0.20)	(0.30, -37.20, -3.30)	(0.00, -37.31, -3.49)	(0.30, 0.11, 0.19)	(0.40, -37.07, -3.20)

Sphere detection - absolute deviations

ETDS QA Overview - L4 | Pads 32°C | Couch 0deg

Lateral (X)



Defl.	Deviation	Lateral (X) [mm]	Longitudinal (Y) [mm]	Vertical (Z) [mm]
D1	SD - Phantom	0.10	0.28	0.01
D1	SD - Surface Tracking	0.10	0.20	0.00
D2	SD - Phantom	0.30	0.11	0.19
D2	SD - Surface Tracking	0.10	0.13	0.10

Absolute deviations of the X-ray based sphere detection from the phantom axis setpoint and from the ETD surface tracking, evaluated at the deflected position (end of measurement window). Tolerance rings: accept 1 mm, watch 2 mm.